Essays

|!|BCEEA0100000001|!| ★★

Describe how the leaves of flowering plants are adapted for photosynthesis, gas exchange and reducing water loss. (11 marks)

##

Adaptations for photosynthesis (Max. 3)

- Leaves are flat and have large surface area to absorb maximum light. (1)
- Leaves are thin to allow light to penetrate to lower mesophyll layers. (1)
- Cuticle and epidermis are transparent to allow light to penetrate to mesophyll. (1)
- Palisade mesophyll cells are packed with chloroplasts to increase photosynthetic capacity / light absorption. (1)
- Xylem is present to supply water to the leaf for photosynthesis. (1)

Adaptations for gas exchange (Max. 3)

- Spongy mesophyll cells are moist / have wet surface / have large surface area for gas exchange. (1)
- Leaves are thin to reduce distance for diffusion of gases. (1)
- Air spaces in spongy mesophyll allow circulation of gases. (1)
- Stomata allow entry and exit of gases, e.g. oxygen, carbon dioxide and water vapour. (1)

Adaptations for reducing water loss (Max. 2)

- Waxy cuticle on upper surface reduces water loss by evaporation. (1)
- More stomata are present in lower epidermis than the upper epidermis to reduce water loss by evaporation. (1)
- Guard cells can close stomata to reduce water loss. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Understanding basic concepts	8
Communication	3

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|!|BCEEA0100000002|!| ★★ **Cross topic**

Humans and land plants absorb nutrients by small intestine and roots respectively. Describe how small intestine and roots are adapted to the absorption of nutrients.

(10 marks)

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Adaptation of small intestine in the absorption of nutrients (Max. 4)

- Small intestine is long and this can provide sufficient time for the absorption of nutrients. (1)
- The inner wall of small intestine is highly folded and has numerous villi. This increases the surface area for absorption of nutrients. (1)
- The epithelial cells have microvilli. This increases the surface area for absorption of nutrients. (1)
- The epithelium is thin that provides a short distance for nutrients to diffuse into the bloodstream. (1)
- The network of blood capillaries allows the absorbed nutrients to be transported away quickly. (1)
- Epithelial cells contain mitochondria that provide energy for the absorption of nutrients (e.g. glucose, amino acids and minerals) by active transport. (1)

Adaptation of roots in the absorption of nutrients (e.g. minerals) (Max. 3)

- Roots have extensive branching that increases the surface area for absorption of minerals. (1)
- There are numerous root hairs developed on the root epidermis. Root hairs increase the surface area for absorption of minerals. (1)
- Root hairs are long and fine. They can penetrate the spaces between soil particles to absorb minerals. (1)
- Root cells contain mitochondria that provide energy for the absorption of minerals by active transport. (1)

Communication (Max. 3)

(10 marks)

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Skills assessed	Marks
Understanding basic concepts	7
Communication	3

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